Notes: Ofek (JFE, 1993)
Capital Structure and Firm Response to Poor Performance

Contents

1 Big picture **3**

2 Incremental contribution of the paper **3**

3 Data and measurement **4**

3.1 Sample construction 4
3.2 Actions during the distress year 4
3.3 Capital structure and ownership variables 5
3.4 Descriptive sample facts 5

4 Methodology and empirical specifications **6**

4.1 Logit regressions for firm responses 6
4.2 Market distress versus industry distress 6
4.3 Debt maturity and debt ownership 7
4.4 Cash-flow classification of operational actions 7
4.5 Managerial and outside ownership tests 7

5 Identification and interpretation **8**

5.1 What the paper can identify 8
5.2 Why predistress leverage matters 8
5.3 Why short-term distress is useful 8
5.4 Debt discipline interpretation 8
5.5 Why management turnover differs 8
5.6 Limitations 8

6 Tables and figures **9**

6.1 Table 1: Sample description, actions, and capital structure 9
6.2 Tables 2 and 3: Main leverage results and market versus industry distress . . 10
6.3 Tables 4 and 5: Debt characteristics and cash-flow channels 11
6.4 Table 6: Type of outside stockholder and operational actions 12

7 Short summary 13

8 Conclusion 14

8.1 Core conclusion . . . 14
8.2 Economic intuition . . . 14
8.3 Incremental contribution revisited . . . 14
8.4 Implications . . . 14
8.5 Limitations . . . 14
8.6 Takeaway . . . 14

1 Big picture

- **Research question.** Does a firm's predistress capital structure affect how quickly and how strongly it responds to a short period of poor performance?
- **Core hypothesis.** Jensen's debt-discipline argument predicts that highly leveraged firms should react faster to declines in firm value, because a smaller value decline can put debtholders at risk and trigger pressure to restructure.
- **Main empirical setting.** The paper studies 358 firms that perform normally or well in a base year and then experience a sharp one-year decline in value, defined as a distress-year stock return in the bottom decile of the market.
- **Main finding.** Higher leverage before distress is strongly associated with operational responses during distress, especially asset restructuring and employee layoffs.
- **Financial responses.** Higher predistress leverage is also associated with debt restructuring, bankruptcy filings, dividend cuts, and larger dividend reductions.
- **Important non-result.** Leverage does not significantly increase management turnover in the first year of distress, suggesting that control transfers to debtholders may be slower for management replacement than for asset or cost restructuring.
- **Market versus industry distress.** The leverage-response relation appears when firms suffer an absolute decline in market value. It is absent for firms that only underperform their industry but do not suffer comparable market distress.
- **Governance result.** Higher managerial ownership reduces the probability of some operational actions, especially actions that do not generate immediate cash inflow. This is consistent with managerial entrenchment.
- **Economic takeaway.** Debt appears to discipline poorly performing firms by pushing them to restructure sooner, but managerial ownership can mute operational adjustment.

2 Incremental contribution of the paper

- **Moves from distress outcomes to distress responses.** Earlier work documents bankruptcy, debt renegotiation, and management turnover in distressed firms. Ofek asks which actions firms take immediately after a short performance shock and how capital structure affects those actions.
- **Focuses on predistress leverage.** The paper measures leverage before the bad year. This helps separate leverage chosen in normal times from leverage mechanically created by accumulated losses.
- **Studies short-term distress.** The sample is designed around a one-year performance decline. This makes the paper especially about speed of response rather than eventual outcomes after years of deterioration.

- **Separates operational from financial actions.** The analysis distinguishes asset restructuring, layoffs, management turnover, debt restructuring, bankruptcy, and dividend cuts.
- **Separates cash-generating from non-cash-generating actions.** This is useful because asset sales may simply raise cash to service debt, while plant closings, discontinued operations, layoffs, and management changes can reflect longer-run value-maximizing restructuring.
- **Contrasts market and industry distress.** The paper shows that leverage disciplines firms mainly when firm value falls absolutely, not merely when the firm performs poorly relative to an industry benchmark.
- **Adds ownership governance evidence.** Managerial holdings and outside blockholders are included to test whether equity ownership structures alter firms' response to distress.

3 Data and measurement

3.1 Sample construction

The sample is built to isolate firms that experience a rapid decline in value after a normal or good year.

- **Sample period.** Base years are 1983 to 1986, and distress years are 1984 to 1987.
- **Universe.** Firms must be publicly traded on NYSE, AMEX, or NASDAQ in both the base and distress years.
- **Size screen.** The market value of equity at the end of the base year must be at least $30 million.
- **Exclusions.** Financial firms and public utilities are excluded.
- **Performance screen.** Firms must be in the top 67% of market returns in the base year and in the bottom 10% of market returns in the distress year.
- **Final sample.** 378 firms satisfy the initial filters, but 20 lack sufficient public data, leaving 358 firms.

Interpretation. The design selects firms that suffer a sudden one-year decline rather than firms that drift into distress over a long period.

3.2 Actions during the distress year

The paper records firm actions using financial statements, the *Wall Street Journal Index*, Compustat, CRSP, proxy statements, and other public sources.

- **Asset restructuring.** Divestitures, asset sales, discontinued operations, or closing/consolidating plants or regional headquarters.
- **Employee layoffs.** A reported layoff or a reduction of at least 10% of the work force, excluding cases where measurement is confounded by major divestitures, asset sales, or acquisitions.
- **Top management replacement.** Replacement of one of the top three managers: chairman, president, or CEO.
- **Debt restructuring.** A creditor agreement after covenant violation or default, or a restructuring or reorganization identified in financial statements.
- **Bankruptcy filing.** Chapter 11 or Chapter 7 filing.
- **Dividend change.** Change in the annual regular dividend from the base year to the distress year.

Interpretation. The first three actions are operational responses; the last three are financial responses.

3.3 Capital structure and ownership variables

The main explanatory variable is predistress leverage:

$$\text{Leverage} = \frac{\text{book value of total debt}}{\text{book value of total debt} + \text{market value of equity}}.$$

Other variables include:

- **Liquidity.** Current assets divided by current liabilities.
- **Firm size.** Log of market value of equity at the end of the base year.
- **Distress-year return.** Annual stock return during the distress year.
- **Debt maturity.** Ratio of long-term debt to total debt.
- **Debt ownership.** Ratio of public debt to total debt; private leverage and public leverage are also considered separately.
- **Managerial holdings.** Share of equity owned by top managers and directors.
- **Outside holdings.** Share of equity owned by the largest nonmanagerial stockholder.

3.4 Descriptive sample facts

- The median base-year stock return is 38.2%, while the median distress-year return is -54.4%.

- 53% of sample firms take some operational action during the distress year.
- 23% undertake asset restructuring, 21% replace top management, and 28% of the measurable layoff sample lay off employees.
- 11% restructure debt, 1% file for bankruptcy, and 47% of dividend-paying firms cut dividends.
- Average leverage rises from 0.20 in the base year to 0.35 at the end of the distress year; the median rises from 0.12 to 0.31.
- Managerial ownership is high in this sample: the mean is 26% and the median is 22%.

4 Methodology and empirical specifications

4.1 Logit regressions for firm responses

For each action, Ofek estimates logit regressions of the form

$$\Pr(Action_i = 1) = F\left(\alpha + \beta Lev_i + \Gamma' X_i\right),$$

where $Action_i$ is an indicator for a response during the distress year, Lev_i is base-year leverage, and X_i includes liquidity, managerial holdings, outside holdings, firm size, and distress-year return.

- A positive β means highly leveraged firms are more likely to take that action.
- The main actions are any operational action, asset restructuring, layoffs, management turnover, debt restructuring, bankruptcy, and dividend cuts.
- The dividend-yield-change regression is estimated by OLS because the dependent variable is continuous.

4.2 Market distress versus industry distress

The paper compares two groups:

- **Market distress.** Firms whose return falls to the bottom 10% of the market after being in the top 67% in the base year.
- **Industry distress only.** Firms whose return falls to the bottom 10% of their three-digit SIC industry after being in the top 67% of that industry, excluding firms also in market distress.

Interpretation. If leverage matters because an absolute decline in value threatens debtholders, leverage should matter more for market distress than for industry-relative underperformance.

4.3 Debt maturity and debt ownership

To study which debt features matter, the paper separates:

- long-term and short-term leverage;
- the ratio of long-term debt to total debt;
- private and public leverage; and
- the ratio of public debt to total debt.

Interpretation. Private debtholders may monitor more actively than dispersed public debtholders, so private debt may be more closely linked to operational response and debt restructuring.

4.4 Cash-flow classification of operational actions

The paper separates operational actions into groups based on their immediate cash-flow consequences.

- **Cash inflow actions.** Asset sales and divestitures that generate current cash.
- **No cash inflow actions.** Discontinued operations, consolidated operations, layoffs, or management changes.
- **No debt-payment actions.** Operational actions that do not raise cash or asset sales whose proceeds are not used to repay debt.

Interpretation. If leverage matters only because firms must raise cash to service debt, it should mainly predict cash-generating actions. If leverage also induces broader discipline, it should predict non-cash operational restructuring.

4.5 Managerial and outside ownership tests

Managerial holdings test entrenchment. If managers with large equity stakes are harder to remove or prefer to maintain firm size, managerial ownership may reduce operational restructuring.

Outside blockholders are split into types: individuals, corporations, corporations friendly to management, banks or insurance companies, investment managers, employee trusts, and investor groups.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies a strong conditional association between predistress leverage and the probability of firm responses to a sharp one-year performance decline. The evidence is consistent with debt discipline, especially because leverage is measured before the distress year.

5.2 Why predistress leverage matters

Using base-year leverage is central. It avoids defining leverage after stock prices fall, when high leverage may simply be a mechanical consequence of distress. Predistress leverage is closer to the capital structure that was in place when the shock arrived.

5.3 Why short-term distress is useful

Short-term distress distinguishes quick responses from eventual responses after years of losses. The low bankruptcy rate in the first year of distress, compared with higher bankruptcy rates in long-distress samples, shows that the timing of distress matters.

5.4 Debt discipline interpretation

The main coefficient should not be read as proof that debt always increases firm value. It says that firms with more debt respond more quickly and more often to poor performance. These responses may preserve going-concern value, but some cash-generating asset sales could also involve distressed-sale costs.

5.5 Why management turnover differs

Leverage predicts asset restructuring and layoffs but not management replacement in the first year. This suggests that debtholders can pressure firms to change assets or costs sooner than they can force a change in top management.

5.6 Limitations

- The design is observational, so leverage may be correlated with unobserved firm characteristics that also affect response.
- The sample is limited to public firms with available data and equity value above $30 million.

- Public sources may miss some actions, especially for smaller firms.
- The paper studies the probability of actions, not their long-run value consequences.
- The period is 1983-1987, so the institutional and bankruptcy environment differs from modern distress settings.

6 Tables and figures

6.1 Table 1: Sample description, actions, and capital structure

Read:

- Panel A verifies that the sample has normal or strong performance in the base year and extremely poor performance in the distress year.
- Panel B documents how common quick responses are: 53% of firms take some operational action in the distress year.
- Asset restructuring occurs in 23% of firms, management replacement in 21%, and debt restructuring in 11%.
- Bankruptcy is rare in the first year of poor performance: only 4 firms, or 1% of the sample, file.
- Panel C shows that leverage rises sharply during distress, while profitability measures fall substantially.
- Panel D shows that large outside stockholders are common, especially investment managers and corporations.

Economic message: The sample is designed around a sharp fall in market value, and firms often react quickly before bankruptcy becomes common.

Table 1

Description of the sample of 358 firms that are in the top 67% of the market in the base year and the bottom 10% in the distress year. The sample period is 1983–1987.

(A) *Sample firms' stock returns in the base and distress years*[a]

	Returns in the base year				Returns in the distress year			
	Low	High	Ave	Med	Low	High	Ave	Med
Total sample	− 15.3	1,940.0	71.2	38.2	− 97.9	− 21.0	− 54.8	− 54.4
Average market return[b]			19.0				13.9	
Base year 1983, distress year 1984	15.9	1,940.0	104.7	55.4	− 85.3	− 49.3	− 60.0	− 58.3
Market return			23.1				5.1	
Base year 1984, distress year 1985	− 15.3	295.7	19.9	3.3	− 94.4	− 21.0	− 38.5	− 36.1
Market return			5.1				30.9	
Base year 1985, distress year 1986	15.9	628.0	89.8	53.4	− 96.3	− 39.2	− 53.7	− 50.8
Market return			30.9				16.9	
Base year 1986, distress year 1987	0.8	736.4	59.6	30.9	− 97.9	− 49.7	− 62.4	− 62.3
Market return			16.9				2.7	

[a]Characteristics of the sample stock returns (in percent) in the year of normal performance (base year) and the immediately following year of poor performance (distress year).
[b]Average market return is the average of the value-weighted market return in the base year or the distress year.

(B) *Distribution of actions taken in the period of poor performance*

Action taken in the distress year	Number of occurrences	Rate of occurrences[a]
Total in sample	358	100%
Operational actions		
Operational action[b]	189	53%
Employee layoffs[c]	74	28%
Asset restructuring[d]	82	23%
Management replacement[e]	76	21%
Asset restructuring with cash inflow[f]	53	15%
Asset restructuring with no cash inflow[g]	44	12%
Financial actions		
Dividend cut[h]	37	47%
Debt restructuring[i]	38	11%
Bankruptcy filings	4	1%

[a]The rate of occurrence is the ratio between the number of occurrences of a certain action and the number of observations in that sample. The number of observations in each sample is 358 except for employee layoffs (264) and dividend cut (78).
[b]Operational actions comprise: asset restructuring, employee layoffs, and management replacement.
[c]Employee layoffs occur if the firm lays off at least 10% of its employees, or if the WSJI reports layoffs.
[d]Asset restructuring occurs if the firm undertakes divestitures, sells assets, or discontinues operations.
[e]Management replacement occurs if the firm replaces one of its top three managers.
[f]Asset restructuring with cash inflow occurs if the firm divests or sells assets in the distress year.
[g]Asset restructuring with no cash inflow occurs if the firm discontinues operations or consolidates operations in the distress year.
[h]Dividend cuts occur if the firm reduces its annual regular dividend in the distress year compared with the base year.
[i]Debt restructuring occurs if the firm restructures its debt following default or to avoid default.

Table 1, panels A-B: Stock returns and actions during distress.

(C) *Sample's capital structure characteristics and profitability*

	Mean	Med	Low	High
Capital structure characteristic				
Market value of equity in base year (millions)	$211	$85	$30	$6,428
Leverage in base year[a]	0.20	0.12	0.00	0.87
Leverage at the end of distress year	0.35	0.31	0.00	0.98
Current ratio in base year[b]	3.80	2.31	0.21	130.22
Current ratio in distress year	3.35	2.07	0.05	104.34
Ratio of long-term to total debt[c]	0.71	0.80	0.00	1.00
Ratio of public debt to total debt[d]	0.12	0.00	0.00	1.00
Equity holdings – Management	0.26	0.22	0.00	0.88
Equity holdings – Outsider	NA	0.06	NA	0.93
Changes in profitability measures[e]				
Change in EBITD[f]	− 0.99[i]	− 0.33[j]	− 109.97	14.75
Change in operating margins[g]	− 0.96[j]	− 0.39[j]	− 59.11	7.92
Change in EBITD/total assets[h]	− 1.00[j]	− 0.41[j]	− 94.25	28.53

[a]Leverage is defined as the ratio of the book value of debt to the sum of the book value of debt and the market value of equity.
[b]Current ratio is defined as the ratio of current assets to current liabilities.
[c]Ratio of book value of debt with maturity of more than one year to total book value of debt.
[d]Ratio of book value of public debt to total book value of debt.
[e]The change in profitability variable X is calculated as $(X_1 - X_0)$/Absolute value (X_0).
[f]EBITD is earnings before interest, taxes, and depreciation.
[g]Operating margins equal the ratio of EBITD in year t to sales in that year.
[h]EBITD/Assets = ratio of EBITD to total book value of assets at year end.
[i]Significant at the 5% level.
[j]Significant at the 1% level.

(D) *Frequency of large nonmanagerial stockholder types in sample firms*[a]

	Number of firms where such stockholder	
Identity of nonmanagerial stockholder	Exists	Is largest investor
Individual	39	29
Corporation	60	56
Corporation friendly to management	14	16
Bank or insurance company	40	29
Mutual fund investment manager	72	56
Employee trust	11	11
Group of investors	14	12

[a]The distribution of nonmanagerial stockholders with at least 5% of the firm's common stock by type. The frequency is provided for firms in which a certain type of nonmanagerial stockholder exists and is the largest nonmanagerial investor.

Table 1, panels C-D: Capital structure, profitability, and outside holders.

6.2 Tables 2 and 3: Main leverage results and market versus industry distress

Read Table 2:

- The dependent variables are indicators for actions taken during the distress year.
- Base-year leverage is positive and highly significant for any operational action.
- Leverage strongly predicts asset restructuring and employee layoffs.
- Leverage does not significantly predict management turnover in the first year.
- Leverage predicts debt restructuring, bankruptcy filing, and dividend cuts.
- Higher managerial holdings significantly reduce management turnover.

Read Table 3:

- Panel A repeats the analysis for firms in market distress only.
- Leverage remains positive and significant for operational actions, asset restructuring, and employee layoffs.

- Panel B studies firms in industry distress only.
- For industry-distress-only firms, leverage is not significantly related to operational actions.
- The contrast suggests that debt discipline is triggered by an absolute decline in firm value, not merely poor relative industry performance.

Economic message: The central result is not simply that weak firms restructure. It is that firms with more debt restructure faster when market value falls enough to threaten debtholder claims.

Table 2

Capital structure and responses to distress for a sample of 358 firms that are in the top 67% of the market in the base year and the bottom 10% in the distress year. The sample period is 1983–1987.[a]

Response	R-sq LR[b]	Constant	Leverage base year[c]	Liquidity base year[d]	Managerial holdings[e]	Outside holdings[f]	Firm size[g]	Return[h]
Operational actions[i]	0.075	– 0.943	2.514[q]	– 0.013	– 0.943	– 1.204	0.285[r]	– 0.711
	36.4[q]	(1.35)	(4.35)	(0.83)	(1.49)	(1.63)	(2.42)	(0.92)
Asset restructuring[j]	0.165	– 3.505[q]	3.295[q]	– 0.180[s]	– 0.470	– 1.580	0.305[r]	1.757[s]
	62.4[q]	(3.85)	(5.01)	(1.73)	(0.59)	(1.61)	(2.20)	(1.79)
Employee layoffs[k]	0.061	– 1.791[r]	1.938[r]	– 0.042	– 1.146	– 1.370	0.153	0.828
	18.5[r]	(1.98)	(2.55)	(0.90)	(1.35)	(1.31)	(1.04)	(0.83)
Management turnover[l]	0.037	– 1.105	– 0.346	0.003	– 1.733[r]	0.917	0.164	– 1.392
	13.4[r]	(1.35)	(0.55)	(0.21)	(2.06)	(1.17)	(1.26)	(1.48)
Debt restructuring[m]	0.183	– 1.303	4.033[q]	– 0.191	– 0.310	– 0.024	– 0.138	– 1.859
	43.2[q]	(1.05)	(5.02)	(1.21)	(0.29)	(0.02)	(0.68)	(1.36)
Bankruptcy filing[n]	0.239	– 5.699[s]	4.865[r]	– 0.225	2.034	2.827	0.238	– 5.277
	10.4	(1.73)	(2.27)	(0.48)	(0.73)	(0.94)	(0.47)	(1.40)
Dividend cut[o]	0.105	– 4.730[q]	1.485[s]	– 0.184	– 0.691	0.793	0.340[r]	2.303[s]
	24.8[q]	(3.99)	(1.91)	(1.28)	(0.63)	(0.79)	(2.04)	(1.88)
Dividend yield change[p]	0.037	0.004[r]	– 0.003[r]	– 0.000	– 0.001	– 0.002	– 0.001[r]	– 0.002
		(2.25)	(2.34)	(0.01)	(0.81)	(0.85)	(1.98)	(1.27)

Table 2: Capital structure and responses to distress.

Table 3

Comparison of responses to poor performance relative to the market (market distress) and relative to the industry (industry distress) for a sample of 358 firms that are in the top 67% of the market (industry) in the base year and the bottom 10% in the distress year. The sample period is 1983–1987.[a]

Responses	R-sq LR	Constant	Leverage base year	Liquidity base year	Managerial holdings	Outside holdings	Firm size	Return distress year
(A) *Capital structure and responses to market distress*								
Operational actions	0.08	– 1.829[c]	2.811[b]	– 0.004	– 0.233	– 0.884	0.459[b]	– 1.391
	29.6[b]	(2.19)	(4.24)	(0.23)	(0.32)	(1.00)	(3.29)	(1.51)
Asset restructuring	0.20	– 4.228[b]	3.809[b]	– 0.213[d]	0.204	– 2.422[c]	0.508[b]	1.000
	58.8[b]	(3.93)	(4.82)	(1.70)	(0.23)	(2.05)	(3.12)	(0.87)
Employee layoffs	0.08	– 1.894[d]	2.285[c]	– 0.033	– 1.633	– 1.864	0.272	– 0.230
	17.2[b]	(1.70)	(2.53)	(0.75)	(1.51)	(1.44)	(1.52)	(0.19)
Management turnover	0.05	– 2.441[c]	– 0.249	0.017	– 0.663	2.323[c]	0.272[d]	– 0.927
	13.0[c]	(2.46)	(0.33)	(0.94)	(0.69)	(2.48)	(1.74)	(0.82)
(B) *Capital structure and responses to industry distress*								
Operational actions	0.24	0.081	1.648	– 0.242	– 0.159	– 2.498	0.610[d]	– 4.690[b]
	18.8[b]	(0.03)	(0.89)	(0.91)	(0.09)	(0.99)	(1.84)	(2.88)
Asset restructuring	0.16	– 1.249	1.688	– 0.301	0.528	– 1.955	0.445	– 3.170[d]
	9.8	(0.50)	(0.97)	(1.03)	(0.28)	(0.63)	(1.32)	(1.93)
Employee layoffs	0.30	8.093[d]	– 2.985	– 0.155	– 6.088	– 4.450	– 0.403	– 9.050[c]
	12.0[d]	(1.70)	(1.11)	(0.47)	(1.38)	(1.23)	(0.84)	(2.41)
Management turnover	0.20	– 2.339	0.424	– 0.179	– 1.691	– 3.683	0.733[c]	– 2.688[d]
	12.2[d]	(0.88)	(0.22)	(0.64)	(0.80)	(0.98)	(2.06)	(1.67)

[a]Coefficients of logistic regressions estimating the relation between the probability of action in the period of poor performance and the capital structure characteristics of the firm for the group of firms with only absolute or only relative poor performance (*t*-statistics in parentheses). Definitions of all terms are given in footnotes b–p of table 2. The number of observations in each regression in panel A is between 261 and 264, except for employee layoffs, with 187; the number of observations in each regression in panel B is 55 or 56, except for employee layoffs, with 40.

[b]Significant at the 1% level.

[c]Significant at the 5% level.

[d]Significant at the 10% level.

Table 3: Market distress versus industry distress.

6.3 Tables 4 and 5: Debt characteristics and cash-flow channels

Read Table 4:

- Both long-term and short-term leverage are positively related to operational actions.
- The long-term debt share itself is not significant, suggesting that the leverage effect is not confined to one maturity category.
- Private leverage is strongly related to operational actions and debt restructuring.
- Public leverage is not significant in the same way, consistent with the idea that private debtholders monitor more actively or renegotiate more easily.
- A higher public-debt share reduces the probability of debt restructuring and management turnover in some specifications.

Read Table 5:

- Leverage predicts asset restructuring with immediate cash inflow.
- Leverage also predicts asset restructuring with no immediate cash inflow.
- Leverage predicts broader operational actions that do not generate current cash, such as layoffs, management changes, and discontinued or consolidated operations.

- Managerial holdings are negatively related to non-cash operational actions.

Economic message: Debt does not merely push firms to sell assets for cash. It also appears to induce cost-cutting and restructuring actions whose benefits come through longer-run operating improvements.

Table 4

Debt characteristics and the firm's actions for a sample of 358 firms that are in the top 67% of the market in the base year and the bottom 10% in the distress year. The sample period is 1983–1987.[a]

Responses	R-sq LR	Constant	Leverage base year	Maturity ratio[b]	Ownership ratio[c]	Long-term leverage[d]	Short-term leverage[d]	Private leverage[e]	Public leverage[e]	Liquidity base year	Firm size
Operational actions	0.067 33.1[f]	− 1.788[f] (3.18)				2.294[f] (3.61)	3.328[h] (1.93)			− 0.013 (0.79)	0.314[f] (2.77)
Operational actions	0.081 36.4[f]	− 1.976[f] (3.10)	2.365[f] (3.65)	0.454 (1.02)						− 0.034 (0.53)	0.298[g] (2.50)
Operational actions	0.075 33.5[f]	− 2.067[f] (3.21)						3.258[f] (4.06)	1.041 (0.88)	0.001 (0.01)	0.348[f] (2.90)
Debt restructuring	0.182 41.4[f]	− 2.645[g] (2.50)						5.119[f] (5.15)	− 0.009 (0.01)	− 0.093 (0.57)	− 0.091 (0.46)
Operational actions	0.073 32.5[f]	− 1.993[f] (3.13)	2.860[f] (4.18)		− 0.554 (1.06)					− 0.007 (0.12)	0.347[f] (2.89)
Debt restructuring	0.193 43.8[f]	− 2.770[f] (2.62)	4.970[f] (5.25)		− 3.260[g] (2.57)					− 0.076 (0.48)	− 0.050 (0.25)
Management turnover	0.028 9.1[h]	− 2.763[f] (3.81)	1.065 (1.48)		− 2.236[g] (2.51)					0.081 (1.21)	0.244[h] (1.82)

[a]Coefficients of logistic regressions estimating the relation between capital-structure characteristics and the probability of actions in the period of poor performance for a sample of firms experiencing one year of distress. These regressions estimate the effect of various debt characteristics on the actions (t-statistics in parentheses). Definitions of all terms are given in footnotes b–p of table 2. The number of observations in each regression is between 318 and 355.
[b]Maturity ratio is the ratio of the book value of long-term debt to total debt, for firms with positive debt in the base year.
[c]Ownership ratio is the ratio of the book value of public debt to total debt, for firms with positive debt in the base year.
[d]Long(short)-term leverage is the ratio of the book value of long(short)-term debt to the book value of total debt plus the market value of equity, in the base year.
[e]Private (public) leverage is the ratio of the book value of nonpublic (public) debt to the book value of total debt plus market value of equity, in the base year.
[f]Significant at the 1% level.
[g]Significant at the 5% level.
[h]Significant at the 10% level.

Table 4: Debt maturity, debt ownership, and firm actions.

Table 5

Short-term cash inflow and capital structure for a sample of 358 firms that are in the top 67% of the market in the base year and the bottom 10% in the distress year. The sample period is 1983–1987.[a]

Responsive actions	R-sq LR[b]	Constant	Leverage base year	Liquidity base year	Managerial holdings	Outside holdings	Firm size	Return distress year
Asset restructuring with cash inflow[b]	0.168 49.4[f]	− 4.785[f] (4.36)	3.425[f] (4.74)	− 0.182 (1.41)	0.956 (1.07)	− 0.220 (0.21)	0.236 (1.49)	2.641[g] (2.30)
Asset restructuring with no cash inflow[c]	0.134 35.4[f]	− 2.893[f] (2.65)	1.952[f] (2.61)	− 0.259[h] (1.74)	− 2.154[g] (1.97)	− 2.433[h] (1.81)	0.300[h] (1.88)	0.814 (0.68)
Operational actions with no cash inflow[d]	0.047 22.8[f]	− 0.729 (1.06)	0.988[h] (1.94)	− 0.013 (0.79)	− 1.620[g] (2.52)	− 1.045 (1.43)	0.264[g] (2.32)	− 0.800 (1.04)
Operational actions with no debt payment[e]	0.049 23.8[f]	− 0.475 (0.69)	1.511[f] (2.89)	− 0.012 (0.78)	− 1.469[g] (2.33)	− 1.275[h] (1.75)	0.204[h] (1.79)	− 0.599 (0.79)

[a]Coefficients of logistic regressions estimating the relation between a firm's capital structure and the probability of operational action in a period of poor performance. The actions are grouped in accordance with the cash flow they generate at the time they are taken (t-statistics in parentheses). Definitions of all terms are given in footnotes b–p of table 2. The number of observations in each regression is between 348 and 352.
[b]Asset restructuring with cash inflow equals 1 if the firm divests or sells assets in the distress year.
[c]Asset restructuring with no cash inflow equals 1 if the firm discontinues operations or consolidates operations in the distress year.
[d]Operational actions with no cash inflow equals 1 if the firm lays off employees or replaces its management or restructures its assets with no cash inflow.
[e]Operational actions with no debt payment equals 1 if the firm takes an operational action with no cash inflow or if the firm divests or sells assets but the proceeds are not used to repay debt.
[f]Significant at the 1% level.
[g]Significant at the 5% level.
[h]Significant at the 10% level.

Table 5: Cash-flow consequences of operational actions.

6.4 Table 6: Type of outside stockholder and operational actions

Read:

- Table 6 asks whether the identity of large outside stockholders predicts operational responses.
- Most outside stockholder types do not significantly increase operational action.
- Investment management ownership is negatively related to operational actions and management turnover in some specifications.
- The results do not support the view that outside blockholders generally force quick operational restructuring in this short-term distress sample.

Economic message: Debt appears to be the stronger disciplining force in this setting. Outside equity blocks, at least as observed in the data, do not systematically push firms to respond to one-year distress.

Table 6

Type of nonmanagerial investor and operational actions for a sample of 358 firms that are in the top 67% of the market in the base year and the bottom 10% in the distress year. The sample period is 1983–1987.[a, b]

Responses	Constant	Leverage base year	Liquidity base year 0	Managerial holdings	*IND*	*CORP*	*FCOR*	*BAIN*	*IVMG*	*EMPT*	*GROU*	Return	Firm size
Operational actions	− 0.853	2.514[c]	− 0.012	− 1.086[e]	0.381	− 0.891	− 2.267	1.090	− 5.730[e]	− 0.014	− 0.877	− 0.795	0.289[d]
	(1.20)	(4.29)	(0.77)	(1.67)	(0.18)	(0.98)	(1.59)	(0.31)	(1.84)	(0.00)	(0.38)	(1.00)	(2.41)
Asset restructuring	− 3.421[c]	3.265[c]	− 0.171[e]	− 0.461	− 3.585	− 1.331	− 1.888	4.613	− 3.132	− 4.939	− 0.276	1.788[e]	0.277[d]
	(3.69)	(4.89)	(1.66)	(0.56)	(1.14)	(1.15)	(0.79)	(1.25)	(0.77)	(0.79)	(0.11)	(1.79)	(1.96)
Employee layoffs	− 1.665[e]	1.833[d]	− 0.042	− 1.210	− 4.254	− 0.201	− 5.713	− 0.264	− 2.060	− 2.304	− 0.769	0.630	0.158
	(1.78)	(2.37)	(0.91)	(1.41)	(0.82)	(0.17)	(1.35)	(0.06)	(0.50)	(0.52)	(0.26)	(0.63)	(1.04)
Management turnover	− 0.883	− 0.391	0.004	2.135[d]	4.627[e]	0.724	− 0.312	− 0.874	− 8.302[e]	1.232	0.716	− 1.664[e]	0.1901
	(1.07)	(0.60)	(0.27)	(2.44)	(1.80)	(0.76)	(0.22)	(0.21)	(1.84)	(0.30)	(0.29)	(1.70)	(1.43)

[a]Coefficients of logistic regressions estimating the relation between capital-structure characteristics and the probability of action in the period of poor performance for a sample of firms experiencing one year of distress. These regressions estimate the effect of the type of the largest nonmanagerial stockholder (t-statistics in parentheses). Definitions of all terms are given in footnotes b–p of table 2.

[b]The following variables equal the actual equity holdings if the type of stockholder owns 0.05 or more of the equity and 0 otherwise. The different types of outside stockholders are: *IND* = individual, *CORP* = a corporation, *FCOR* = a corporation that is managed or controlled by the firm's management, *BAIN* = a bank or an insurance company, *IVMG* = a mutual fund, pension fund, or other investment management firm, *EMPT* = employee trust, and *GROU* = a group of related investors.

[c]Significant at the 1% level.

[d]Significant at the 5% level.

[e]Significant at the 10% level.

Table 6: Nonmanagerial investor type and operational actions.

7 Short summary

- The paper studies whether predistress leverage affects firm responses to a sudden year of poor performance.
- The sample contains 358 public firms from 1983-1987 that move from the top 67% of market returns in the base year to the bottom 10% in the distress year.
- Firms react quickly: 53% take an operational action in the distress year.
- Higher predistress leverage strongly predicts operational actions.
- The strongest operational responses are asset restructuring and employee layoffs.
- Leverage does not significantly predict management replacement in the first year of distress.
- Higher leverage also predicts debt restructuring, bankruptcy filing, dividend cuts, and larger dividend reductions.
- The leverage effect appears for market distress but not for industry-relative distress only.
- Both long-term and short-term debt matter; private debt appears more strongly related to actions than public debt.
- Leverage predicts both cash-generating restructuring and non-cash operational restructuring.
- Higher managerial holdings reduce the probability of operational actions that do not generate immediate cash, consistent with entrenchment.
- Outside blockholders generally do not increase the probability of operational responses.

8 Conclusion

8.1 Core conclusion

Ofek shows that capital structure matters for how firms respond to poor performance. Firms with more debt before distress are more likely to restructure operations and financial claims after a sharp decline in value.

8.2 Economic intuition

Debt changes the timing of discipline. When leverage is high, a smaller decline in firm value can threaten default and bring debtholders into the picture. This pressure can push firms to sell assets, cut costs, lay off employees, restructure debt, and reduce payouts sooner than they otherwise would.

8.3 Incremental contribution revisited

The paper's key contribution is to show that leverage affects the speed and type of response to short-term distress. It does not only study firms after prolonged distress or bankruptcy. It shows that the disciplining role of debt appears already in the first year of poor performance.

8.4 Implications

The findings support the view that debt can preserve going-concern value by forcing quicker action. They also imply that low leverage may allow poorly performing firms to delay difficult restructuring. However, debt discipline is not costless: leverage also increases the probability of financial distress and bankruptcy.

8.5 Limitations

The paper cannot fully prove that leverage causes value-improving responses, because capital structure is not randomly assigned. It also does not directly measure whether each restructuring action increases value. The evidence is strongest as a reduced-form link between predistress leverage and the probability of quick response.

8.6 Takeaway

The enduring takeaway is that debt is a governance mechanism. In Ofek's sample, high predistress leverage makes firms respond faster to poor performance, while high managerial ownership can make managers less willing to take painful operational actions.